Algo Consistency Index (ACI)

A daily, pre-registered reading of whether the market is paying the patterns that retail FX and CFD algos trade

Methodology v0.2 · first public print 2026-10-26 · https://thecryptocrat.github.io/algo-consistency-index/ · prepared for MOL, 26 October 2026

What it does

Every trading algo is a pattern bet. A breakout system needs follow-through after a range is broken; a mean-reversion system needs ranges that hold; a session-clock system needs the day's shape to repeat at the same hours. When price action turns erratic, those patterns stop paying long before any single account's statistics can show it.

The ACI measures that directly. Each day it replays a fixed basket of textbook recipes, one family at a time, on real tick data with real trading costs, and reports whether the market has been paying each family: relative to that family's own two-year history, and relative to random entries that use the same exits. The three families are trend/breakout (TB), mean-reversion (MR) and session-clock (SC).

It is published as three numbers (one per family), a two-year chart, three sentences on what moved in the market's structure, a scoreboard of production accounts, and the full validation record, with failures included, on one public page with JSON and CSV feeds.

Why account P&L alone cannot do this

Telling a strategy whose edge has gone from one in an ordinary drawdown takes, in round numbers, $1/S^2$ years of returns, where S is the annual Sharpe ratio: about a year for a Sharpe-1 strategy and four years for a Sharpe-0.5 one. Judging algos by their recent P&L is therefore lagging by construction, and ranking them by recent P&L selects noise. The ACI reads the state of the market instead, so it can move before the account statistics do, and it keeps P&L in its proper place: as a coincident scoreboard and as kill lines, never as the sizing signal.

How it works: three layers on one backbone

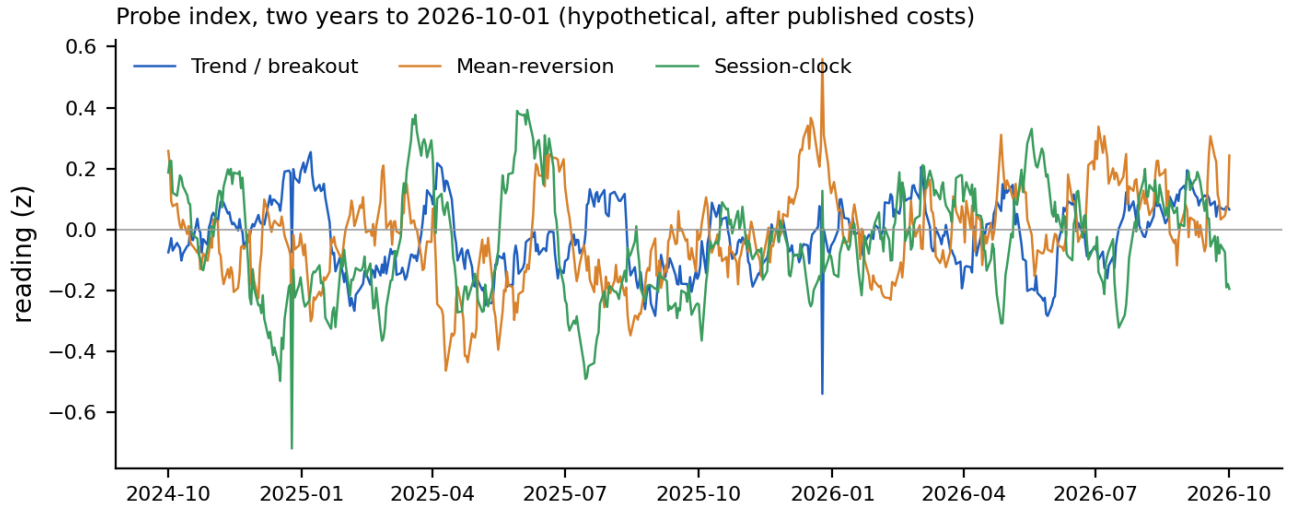
The probe basket (backbone). Eleven frozen textbook recipes, traded long and short, on 17 symbols: gold, US500, US30, USTECH, RUSS2000, EURUSD, USDJPY, AUDUSD, AUDJPY, GBPJPY, USDCHF, AUDNZD, NZDUSD, USDCAD, GBPCHF, GBPUSD and ETHUSD. Trend/breakout: Donchian daily, London and New York opening-range breakouts, 15-minute range break, EMA cross. Mean-reversion: RSI(2), band fade, prior-day level fade, Asian-range fade. Session-clock: fixed-time entry, post-fix drift, pre-rollover fade. A random-entry control with the same exit rules (two geometries, 50 seeds) runs beside them. Exits are walked tick by tick. Costs are the broker's published spread plus the raw spread's excess over its yearly median plus $4 per lot round-turn commission, so every probe is cost-negative on average, as intended: the reading is relative.

The reading. For each family, the 20-day net R per trade of its probes minus the random control's, expressed as a z-score against the family's own 504-day history. A reading near zero means the market is paying that family about as it usually does; the public cards show the reading's percentile within the last two years (a card reading 74 means better than 74 percent of the last two years for that family).

Layer 1, Scoreboard (coincident). For production accounts tagged to each family: the breadth of accounts positive over 20 days, a 20-day momentum z-score with day-clustered errors, realised drawdown against each algo's budget, and a sequential-probability (Wald SPRT) kill count per algo. It is labelled lagging on the page, because it is.

Layer 2, Pattern Integrity (leading candidates). Fifteen market-structure features per symbol (volatility ratio, volatility of volatility, gap and range against ATR, efficiency ratio, 15-minute autocorrelation, variance ratio, breakout follow-through rate, mean-reversion hit rate, session-shape fidelity, time-of-high and time-of-low deviation, spread and tick stress, event density), each rank-normalised against 2021-2024, plus a Regime Drift distance of the current ten-day state from that reference window. The page turns the largest moves into three plain sentences.

Layer 3, Nowcast and sizing (withheld). A walk-forward model would turn Layer 2 into a 0-100 nowcast and a banded sizing multiplier between 0.5 and 1.5 per family, with a three-day dwell and at most two changes a month, applied at book level, never by ranking sleeves. It is published only once it passes its tests. It has not yet, so the page says so and no multiplier is shown.



Probe index, two years. Hypothetical results of frozen recipes replayed on historical ticks with published costs; not investment performance. Readings move when the tick data is refreshed; the scoreboard moves daily.

Validation ladder and the verdicts so far

Every test is written down before it runs, with its pass bar, the statistical power the data can give it, and a shift-control twin (the same statistic computed 40 days out of alignment) that must fail. Failures are published. This is what keeps the ACI from being a story told after the fact.

Test	Question	Verdict (26 Oct 2026)
T0	Do the market features co-move with live production P&L;?	Not passed at low power (3-7 twenty-day blocks per family)
T1	Does the probe index track live accounts of the same family? (rank corr. >= 0.40 at 20 days, interval above zero, twin fails)	TB passed at low power: rho 0.67, 90% interval 0.37 to 0.82, 7 blocks, twin -0.16. MR and SC not passed.
T2	Can Layer 2 forecast the next 5-10 days of family return out of sample? (two folds, 2026 sealed, twins and permuted labels must fail)	Not passed: twins met the bar in 19 of 48 cells, real features in 14. Layer 3 withheld.
T3	Does the sizing rule beat constant sizing at matched realised drawdown?	Not run (consequence of T2).
T4	Three months of live daily prints	Started 26 Oct 2026 with the first public print.

Read plainly: today the ACI is a validated descriptive instrument for trend/breakout, a descriptive instrument still awaiting its external test for mean-reversion and session-clock, and not yet a prescriptive one. It will not size anything until a nowcast passes on a new sealed period.

How to read the page

- **Cards.** One per family: the reading's percentile within its own two years (today TB 74, MR 95, SC 22), the z-score, the number of probe legs and trades behind it, and the date of the readings. The fourth card is the cost regime: spread plus commission as a share of ATR-sized risk.
- **Chart.** Two years of readings per family; the backfill (computed after the fact) is dashed, the live record is solid.
- **What moved.** The three largest Layer 2 moves over five days, in words, with the family each one weighs on.
- **Scoreboard.** Breadth, momentum, drawdown load and kill count per family for production accounts; shown only when at least five accounts are live in a family.
- **Validation record.** Each gate and its current verdict, updated whenever a test runs.
- **Feeds.** data/aci_latest.json and data/aci_history.csv at https://thecryptocrat.github.io/algo-consistency-index/ for the MOL website or any dashboard.

Benefits

- **For the trader running a book:** a mechanism-specific weather report. When the TB reading sits in its bottom decile, breakout recipes are failing after costs across 17 symbols, not just in one account, and the question changes from "is my algo broken?" to "is the market paying this pattern?".
- **For sizing, once validated:** the multiplier is driven by market state, not by recent P&L, so it cannot become a performance-chasing device, and it acts on the book per family rather than ranking sleeves, which the research record shows selects cost and noise.
- **For investors and clients:** transparency that is rare in this space. The method is frozen and pre-registered, every test's power and result is on the page, failures included, and probe results are labelled hypothetical because they are.
- **For MOL:** a public instrument that nothing published matches (the CTA trend indices are licensed, lag a day and ignore retail costs), a daily reason to visit the site, and context for the demo fleet's results.
- **For research:** it separates "the algo broke" from "the market stopped paying that pattern", which is the first question every post-mortem asks and the one account statistics answer last.

Limits and what comes next

- Probe and Layer 2 readings move only when the tick data is refreshed (monthly so far); the Scoreboard moves daily. A daily tick refresh turns the page into a true daily print.
- The live panel is small (seven 20-day blocks of overlap). An external panel of several hundred live DARWINs with daily quotes and attribute scores is the critical path to settle T1 for mean-reversion and session-clock; the access token exists and the pre-registered test is written, and the pull starts as soon as the data host answers.
- No sizing is published until a nowcast passes T2 and T3 on a new sealed period (2027 H1). 2026 stays sealed as the live record.

Not investment advice. Probe results are hypothetical replays of fixed recipes on historical data with published costs and are not the performance of any account. Multipliers, when published, are relative to the reader's own risk.